

Henri Stewart Wilks-Stebbins

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Doctoral Researcher and Earth Scientist specialising in the application of Deep Learning and High-Performance Computing to quantify landscape sensitivity. Currently investigating the coupled bio-geo-responses of the Eastern Alps to climate change by synthesising Big Data remote sensing time series and climate model data. Expert in developing automated Python-based workflows, as well as operations within Google Earth Engine to establish non-linear correlations between climate variables, biotic factors, and geomorphological failure states.

Education

Ph.D. (dr.rer.nat) in Geology

University of Salzburg, Austria

expected Mar. 2029

M.Sc. in Earth Sciences

Vrije Universiteit Amsterdam, Netherlands

Jun. 2025

B.Sc. in Physical Geography

Liverpool Hope University, England

Jun. 2023

Experience

Doctoral Researcher

University of Salzburg, Division of Geology and Physical Geography

Salzburg, Austria

Mar. 2029

- Investigating the influence of climate change on landscape evolution in the Eastern Alps using Big Data workflows to facilitate large-scale space-time analyses

Research Assistant, Geomorphology and Quaternary Sedimentology Group

Vrije Universiteit Amsterdam, Department of Earth Science

Amsterdam, Netherlands

Aug. 2024 – Apr. 2025

- Leading remote-sensing data acquisition for post-glacial landslide research in New Zealand
- Collaborating within a global, multi-disciplinary team of scientists (New Zealand, UK, Netherlands)

Skills

Geospatial Data Science & AI: Deep Learning and ML, Cloud and Cluster Computing, Big Data Processing

Programming Languages: Python, Bash, JavaScript, MATLAB, Linux

Analytical & Field: DEM analysis, Frequency-Magnitude Scaling, Landform Sensitivity Modelling, Hydro-Biotic Modelling, Geomorphological Mapping, XRF, Remote Sensing

Communication: Doctoral Representative, Scientific Technical Writing, Oral Presentation, Organising Workshops

Publications

Towards a Stress-Based Stability Criterion for Rock Slopes From 3D Stress Modeling of Entire Mountain Massifs (Haunsperger, V., Robl, J., Hergarten, S., Argentin, A.L., Wilks-Stebbins, H., Schröder, A., in prep)

Hidden Erosion: Quantifying Carbonate Mass Loss in the Tauern Catchment, Northern Calcareous Alps (Pleyer, B., Robl, J., Dremel, F., Wilks-Stebbins, H., Tenczer, V., Otto, J.C., Hilberg, S., Rinder, T., Hauzenberger, C., in prep)

Tectonic and Climatic Controls on Valley-Floor Width Along the Alpine Fault, New Zealand (Clubb, F.J., Roda-Boluda, D.C., Wilks-Stebbins, H. and de Vries, L., *American Geophysical Union Fall Meeting 2024*, oral presentation by F. Clubb, and in prep)

Past and Ongoing Research Projects

Landscape Response to a Changing Climate in the Eastern Alps: Analysing Big Data for Past, Present, and Future Conditions
(PhD Dissertation)

Oct. 2025 – Present

- Computing spatio-temporal gradients in space and time across the Eastern Alps
- Determining changes in the erosional potential of Alpine torrents
- Employing supervised deep learning to detect hillslopes close to critical state failure

Constraining Landslide Rates, Controls, and Spatial Distributions in Calabria, Southern Italy (MSc Thesis)

Jun. 2025

- Student collaborator through MSc Thesis on the RELICT project: examining how sediment dynamics and inherited relief impact landscape response to tectonic perturbations
- My role in this project involved creating multi-year landslide inventories which refined hazard assessment and landslide erosion rate estimates

Landslide Frequency and Magnitude in the Southern Alps, New Zealand (MSc Research Project)

Dec. 2024

- Independent 7-month research project delivering a novel multi-year landslide inventory with >2,700 landslides
- Implemented advanced geomorphological and numerical modelling techniques to analyse landslide frequency–magnitude trends
- Co-author of a talk presented at the American Geophysical Union Fall Meeting 2024 and of the future publication in preparation, to which the landslide data I generated is incorporated

The Implications of Climate Change on High Geomorphology and Paraglacial Slope Failures (BSc Thesis)

Jun. 2023

- Investigated the impact of climate on hillslope processes, analysing mass movement frequency–magnitude
- Assessed geological properties, slope mechanics, and climate forcing to identify key drivers of instability
- Conducted systematic literature review to synthesise current research on slope failure mechanisms

Professional Development

Member, Doctoral Mentoring Committee (AlpsChange)

Jan. 2026 – Present

- Co-managing a gender-balanced advocacy body to support 20+ PhD students in conflict resolution and career planning
- Serving as a primary point of contact for students to address institutional or supervisory challenges, ensuring impartial peer support and program transparency

Landslide Mapping „Hackathon“

May. 2025

- Organised and led a one-day landslide mapping event for undergraduate students
- Delivered hands-on instruction in landslide delineation techniques and volume-area scaling calculations
- Served as a primary point of contact for participants, providing technical guidance and support throughout the event

Awards

- Awarded the **Liverpool Geological Society Prize** as Best Performing Liverpool Hope Student in Physical Geography, 2023.
- First Class Honours Bachelor of Science Degree (**Summa Cum Laude**)

References

Dr. Jörg Robl

Associate Professor, Universität Salzburg
joerg.robl@plus.ac.at

Dr. Duna Roda-Boluda (has recently returned to University of Barcelona)

Assistant Professor, Vrije Universiteit
d.c.rodaboluda@vu.nl

Dr. Benjamin Campforts

Assistant Professor, Vrije Universiteit
b.campforts@vu.nl